

Exploratory analysis of ridership and weather

Group 15

Wednesday 17th December, 2025



```
import os
os.chdir("../")

import pandas as pd
import numpy as np

# - - - Load datasets - - -
from src.data_utils import load_weather_data

daily = load_weather_data("data/daily_stats.csv")
weather_station = load_weather_data("data/USW00023272.csv")
weather = load_weather_data("data/weather_2018_2024.csv")

# - - - Clean weather_station data - - -
weather_station = weather_station[["DATE", "PRCP", "TMAX", "TMIN"]].copy()
weather_station["DATE"] = pd.to_datetime(weather_station["DATE"])

for col in ["PRCP", "TMAX", "TMIN"]:
    weather_station[col] = (
        weather_station[col]
        .astype(str)
        .str.strip()
        .replace({"M": np.nan, "": np.nan, "T": "0.0"})
    )
    weather_station[col] = pd.to_numeric(weather_station[col], errors="coerce")

# Convert from tenths of °C/mm to normal units
weather_station["TMAX"] = weather_station["TMAX"] / 10
weather_station["TMIN"] = weather_station["TMIN"] / 10
weather_station["PRCP"] = weather_station["PRCP"] / 10
weather_station["TAVG"] = (weather_station["TMAX"] + weather_station["TMIN"]) / 2

# - - - Clean daily rentals data - - -
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daily["date"] = pd.to_datetime(daily["date"])

# - - - Clean weather_2018-2024 data - - -
weather["DATE"] = pd.to_datetime(weather["DATE"], errors="coerce")

# - - - Quick checks - - -
print("Daily stats sample:")
print(daily.head(), "\n")
print("Weather station sample:")
print(weather_station.head(), "\n")
print("Weather 2018-2024 sample:")
print(weather.head())

/home/jovyan/final -group15/src/data_utils.py:21: DtypeWarning: Columns (17,19,21,23,25,27,2
return pd.read_csv(path)

```

Daily stats sample:

	date	avg_duration_sec	pct_bike_share_yes	pct_subscriber \
0	2018 -01 -01	1870.569455	9.818182	57.818182
1	2018 -01 -02	979.447417	5.965560	88.407134
2	2018 -01 -03	739.360168	4.865243	91.774589
3	2018 -01 -04	730.779091	5.000000	91.606061
4	2018 -01 -05	756.418605	5.953488	88.930233

	total_trips
0	1375
1	3252
2	2857
3	3300
4	2150

Weather station sample:

	DATE	PRCP	TMAX	TMIN	TAVG
0	1921 -01 -01	0.0	15.6	7.2	11.40
1	1921 -01 -02	0.0	12.2	7.8	10.00
2	1921 -01 -03	0.0	15.6	10.0	12.80
3	1921 -01 -04	0.0	16.1	8.3	12.20
4	1921 -01 -05	9.7	12.2	8.3	10.25

Weather 2018-2024 sample:

	STATION	DATE	LATITUDE	LONGITUDE	ELEVATION \
0	USW00023272	2018 -01 -01	37.7705	-122.4269	45.7
1	USW00023272	2018 -01 -02	37.7705	-122.4269	45.7
2	USW00023272	2018 -01 -03	37.7705	-122.4269	45.7
3	USW00023272	2018 -01 -04	37.7705	-122.4269	45.7
4	USW00023272	2018 -01 -05	37.7705	-122.4269	45.7

		NAME	PRCP	PRCP_ATTRIBUTES	SNOW	SNOW_ATTRIBUTES	\
0	SAN FRANCISCO DOWNTOWN, CA US	0.0	,,W,2400	NaN	NaN		
1	SAN FRANCISCO DOWNTOWN, CA US	0.0	,,W,2400	NaN	NaN		
2	SAN FRANCISCO DOWNTOWN, CA US	23.0	,,W,2400	NaN	NaN		
3	SAN FRANCISCO DOWNTOWN, CA US	15.0	,,W,2400	NaN	NaN		
4	SAN FRANCISCO DOWNTOWN, CA US	66.0	,,W,2400	NaN	NaN		

	...	WT05	WT05_ATTRIBUTES	WT08	WT08_ATTRIBUTES	WT11	WT11_ATTRIBUTES	\
0	...	NaN	NaN	NaN	NaN	NaN	NaN	
1	...	NaN	NaN	NaN	NaN	NaN	NaN	
2	...	NaN	NaN	NaN	NaN	NaN	NaN	
3	...	NaN	NaN	NaN	NaN	NaN	NaN	
4	...	NaN	NaN	NaN	NaN	NaN	NaN	

	WT16	WT16_ATTRIBUTES	WT18	WT18_ATTRIBUTES
0	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN

[5 rows x 52 columns]

```
import seaborn as sns
import matplotlib.pyplot as plt

# Merge bike rentals with cleaned weather data
merged = pd.merge(
    daily,
    weather_station,
    left_on="date",
    right_on="DATE",
    how="inner"
)

print("Merged data sample:")
print(merged.head())
print("\nMerged shape:", merged.shape)

# Quick correlation check between weather and rentals
corr_vars = ["total_trips", "TMAX", "TMIN", "TAVG", "PRCP"]
print("\nCorrelation matrix:")
print(merged[corr_vars].corr())

# Visualization Temperature vs Rentals
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sns.scatterplot(data=merged, x="TAVG", y="total_trips", alpha=0.6)
plt.title("Bike Rentals vs Average Temperature (°C)")
plt.show()

# Visualization Rain vs Rentals
merged["Rainy"] = merged["PRCP"] > 0
sns.boxplot(data=merged, x="Rainy", y="total_trips")
plt.title("Bike Rentals on Rainy vs Dry Days")
plt.xlabel("Rainy Day?")
plt.ylabel("Total Trips")
plt.show()

# Visualization monthly trends
merged["month"] = merged["date"].dt.month
sns.lineplot(data=merged, x="month", y="total_trips", estimator="mean")
plt.title("Average Monthly Bike Rentals")
plt.xlabel("Month")
plt.ylabel("Average Trips")
plt.show()

# Visualization Rentals by Precipitation Level
bins = [0, 0.1, 2, 10, 50, merged["PRCP"].max()]
labels = ["0 mm (Dry)", "0.1-2 mm (Light)", "2-10 mm (Moderate)",
          "10-50 mm (Heavy)", "50+ mm (Extreme)"]

merged["prcp_bucket"] = pd.cut(merged["PRCP"], bins=bins, labels=labels, include_lowest=True)

merged["year"] = merged["date"].dt.year

plt.figure(figsize=(10,5))
sns.barplot(data=merged, x="prcp_bucket", y="total_trips", estimator="mean")
plt.title("Average Bike Rentals by Precipitation Level")
plt.xlabel("Precipitation Category")
plt.ylabel("Average Total Trips")
plt.xticks(rotation=30)
plt.show()

# Visualization Rentals by Year and Precipitation Level 2018 -2024
plt.figure(figsize=(12,6))
sns.lineplot(
    data=merged,
    x="year",
    y="total_trips",
    hue="prcp_bucket",
    estimator="mean"

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)
plt.title("Average Bike Trips by Precipitation Level, by Year")
plt.xlabel("Year")
plt.ylabel("Average Trips")
plt.legend(title="Precipitation Bucket")
plt.show()

```

Merged data sample:

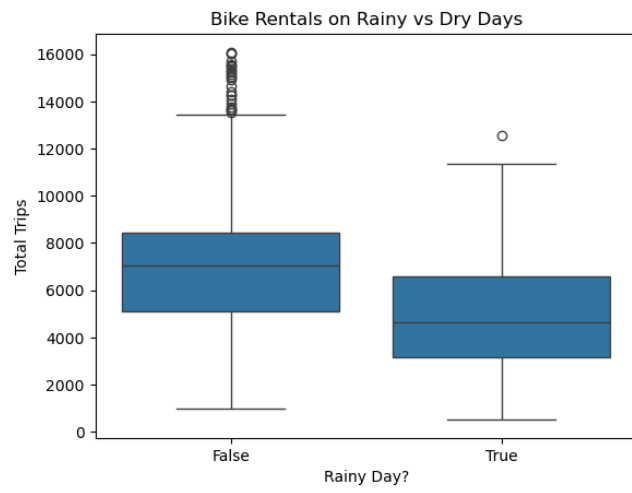
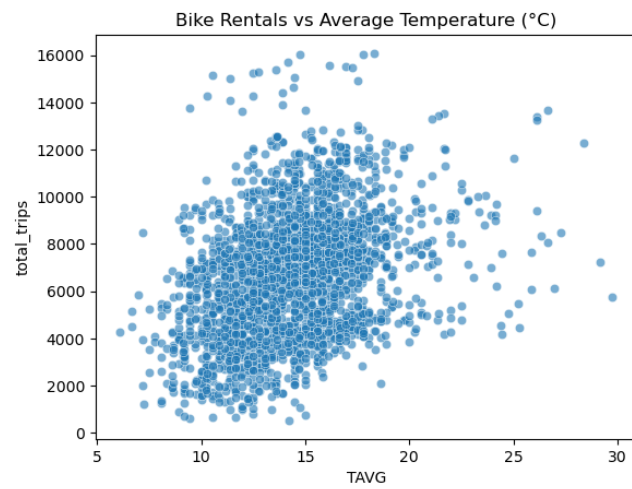
	date	avg_duration_sec	pct_bike_share_yes	pct_subscriber	\
0	2018 -01 -01	1870.569455	9.818182	57.818182	
1	2018 -01 -02	979.447417	5.965560	88.407134	
2	2018 -01 -03	739.360168	4.865243	91.774589	
3	2018 -01 -04	730.779091	5.000000	91.606061	
4	2018 -01 -05	756.418605	5.953488	88.930233	

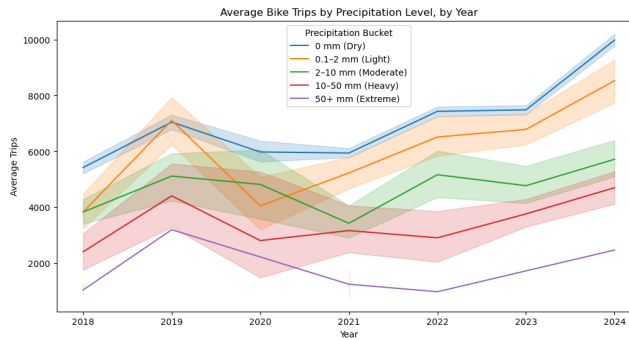
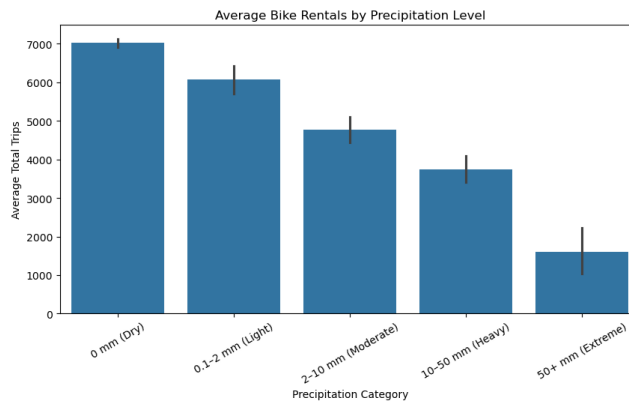
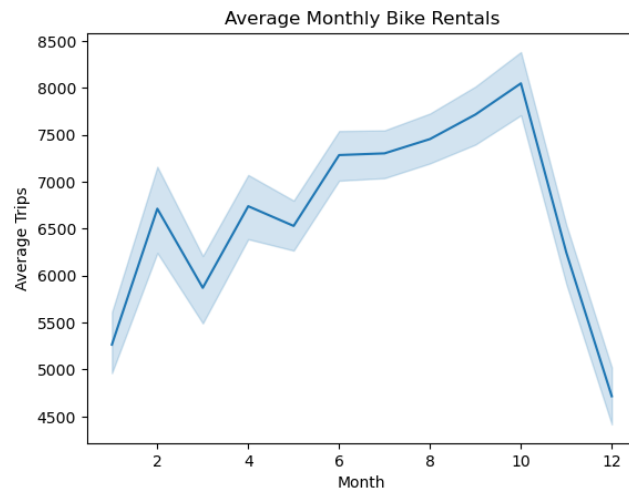
	total_trips	DATE	PRCP	TMAX	TMIN	TAVG
0	1375	2018 -01 -01	0.0	16.1	8.9	12.50
1	3252	2018 -01 -02	0.0	16.1	11.1	13.60
2	2857	2018 -01 -03	2.3	14.4	11.7	13.05
3	3300	2018 -01 -04	1.5	17.2	11.7	14.45
4	2150	2018 -01 -05	6.6	16.1	11.1	13.60

Merged shape: (2528, 10)

Correlation matrix:

	total_trips	TMAX	TMIN	TAVG	PRCP
total_trips	1.000000	0.352185	0.298489	0.355778	-0.275388
TMAX	0.352185	1.000000	0.715201	0.957751	-0.198440
TMIN	0.298489	0.715201	1.000000	0.885992	-0.109690
TAVG	0.355778	0.957751	0.885992	1.000000	-0.176770
PRCP	-0.275388	-0.198440	-0.109690	-0.176770	1.000000





1.1 Correlation Matrix Analysis:

The correlation matrix indicates that temperature has a moderate positive correlation with daily bike rentals ($r \approx 0.35$). This means warmer days generally encourage more biking activity. Precipitation shows a negative correlation ($r \approx -0.27$), confirming that rainfall suppresses demand. These correlations are statistically meaningful, though not perfect, implying that other behavioral and temporal factors also influence usage.

1.2 Bike Rentals vs. Average Temperature Analysis:

The scatterplot reveals a clear upward trend: as average temperature increases, total bike rentals also rise. Most data points cluster between 10–17 °C, reflecting San Francisco’s mild maritime climate. Even within this narrow temperature band, warmer days correspond with noticeably higher ridership. This suggests that modest temperature shifts within SF’s stable climate still have tangible effects on biking behavior.

1.3 Bike Rentals on Rainy vs. Dry Days Analysis:

The boxplot shows a pronounced difference between rainy and dry days. Median ridership drops significantly on rainy days, roughly 1,500 fewer trips per day, according to the regression results. This confirms that rain has an immediate and strong dampening effect on biking, likely due to safety and comfort concerns.

1.4 Average Monthly Bike Rentals Analysis:

The monthly trend line captures a clear seasonal pattern. Ridership rises through the spring and summer, peaking in September-October, before falling sharply in winter. This seasonal cycle mirrors weather conditions and daylight availability, but it also aligns with commuting and recreational habits tied to the academic and work year.

1.5 Average Bike Rentals by Precipitation Level Analysis:

The precipitation levels show a clear descending pattern in bike rentals. As precipitation levels increase in mm, bike rentals decrease as expected. This can be caused by the increased level of danger of biking as the ground gets increasingly wet.

1.6 Average Bike Trips by Year:

This chart shows that bike usage is consistently highest on dry days, with trips steadily increasing from 2018 to 2024. I was curious to see if there was an

increase during 2021 with the peak of COVID and people trying to do outdoor activities, but it actually led to a decrease of bike rentals.

2

```
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

# 1) Build a clean regression frame
reg = merged[["total_trips", "TAVG", "PRCP", "date"]].copy()

reg = reg.replace([np.inf, -np.inf], np.nan).dropna(subset=["total_trips", "TAVG", "PRCP"])

# 2) Simple OLS with robust SEs
X = sm.add_constant(reg[["TAVG", "PRCP"]])
y = reg["total_trips"]
model = sm.OLS(y, X).fit(cov_type="HC1")
print(model.summary())

# 3) Add day -of -week controls and a rainy -day flag
reg["dow"] = reg["date"].dt.dayofweek          # 0=Mon ... 6=Sun
reg["rain"] = (reg["PRCP"] > 0).astype(int)

model_dow = smf.ols("total_trips ~ TAVG + PRCP + C(dow)", data=reg).fit(cov_type="HC1")
print("\n=== With day -of -week controls ===")
print(model_dow.summary().tables[1])

model_rain = smf.ols("total_trips ~ TAVG + rain + C(dow)", data=reg).fit(cov_type="HC1")
print("\n=== Using rainy -day indicator instead of PRCP (mm) ===")
print(model_rain.summary().tables[1])
```

OLS Regression Results

```
=====
Dep. Variable:          total_trips    R -squared:                0.173
Model:                  OLS           Adj. R -squared:            0.172
Method:                 Least Squares  F -statistic:              203.6
Date:                  Tue, 16 Dec 2025  Prob (F -statistic):        1.21e -82
Time:                  07:54:27        Log -Likelihood:           -23024.
No. Observations:      2507           AIC:                      4.605e+04
Df Residuals:          2504           BIC:                      4.607e+04
Df Model:               2
Covariance Type:        HC1
=====
```

	coef	std err	z	P> z	[0.025	0.975]
--	------	---------	---	------	--------	--------

```

-----
const      2831.2290    255.464    11.083    0.000    2330.529    3331.929
TAVG       274.0104     17.311    15.829    0.000     240.081    307.940
PRCP       -89.9538     11.702    -7.687    0.000    -112.889   -67.019
=====
Omnibus:                99.330    Durbin -Watson:                0.359
Prob(Omnibus):          0.000    Jarque -Bera (JB):            115.369
Skew:                   0.462    Prob(JB):                     8.87e -26
Kurtosis:               3.502    Cond. No.                     74.3
=====

```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

=== With day -of -week controls ===

```

=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept    2602.9735    269.361     9.664     0.000    2075.035    3130.912
C(dow) [T.1]   717.3817    167.520     4.282     0.000     389.049    1045.714
C(dow) [T.2]   817.1166    169.062     4.833     0.000     485.762    1148.471
C(dow) [T.3]   887.8696    172.277     5.154     0.000     550.213    1225.526
C(dow) [T.4]   698.9592    160.574     4.353     0.000     384.240    1013.679
C(dow) [T.5]  -202.9581    162.266     -1.251     0.211    -520.994     115.078
C(dow) [T.6]  -929.0850    154.271     -6.022     0.000   -1231.450    -626.720
TAVG          270.2116     16.886    16.002     0.000     237.116     303.308
PRCP          -90.2503     12.637     -7.142     0.000    -115.019    -65.481
=====

```

=== Using rainy -day indicator instead of PRCP (mm) ===

```

=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept    3144.4503    277.368    11.337     0.000    2600.819    3688.081
C(dow) [T.1]   708.1928    170.070     4.164     0.000     374.862    1041.524
C(dow) [T.2]   809.5919    169.815     4.767     0.000     476.760    1142.423
C(dow) [T.3]   870.3451    173.107     5.028     0.000     531.061    1209.629
C(dow) [T.4]   692.7806    161.741     4.283     0.000     375.774    1009.787
C(dow) [T.5]  -202.8711    162.751     -1.247     0.213    -521.857     116.115
C(dow) [T.6]  -962.0571    154.666     -6.220     0.000   -1265.197    -658.917
TAVG          242.2694     17.152    14.124     0.000     208.651     275.888
rain         -1480.8468    122.382    -12.100     0.000   -1720.711   -1240.982
=====

```

```

# Predictions using the rainy -day model on the same cleaned frame
reg["predicted"] = model_rain.predict(reg)

```

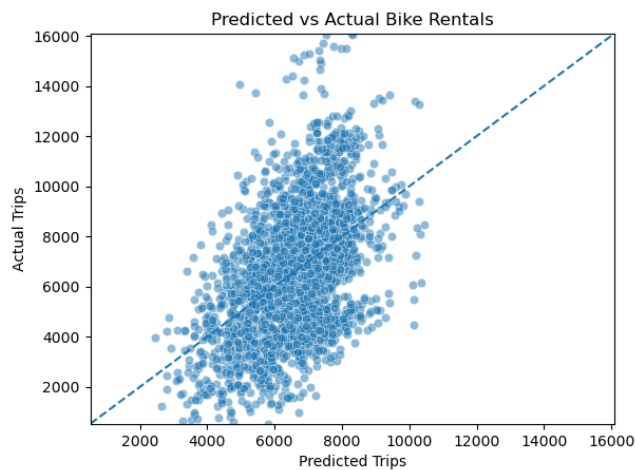
```

import seaborn as sns
import matplotlib.pyplot as plt

sns.scatterplot(data=reg, x="predicted", y="total_trips", alpha=0.5)
plt.xlabel("Predicted Trips")
plt.ylabel("Actual Trips")
plt.title("Predicted vs Actual Bike Rentals")

# 45 -degree line
lims = [min(reg["predicted"].min(), reg["total_trips"].min()),
        max(reg["predicted"].max(), reg["total_trips"].max())]
plt.plot(lims, lims, linestyle="--")
plt.xlim(lims); plt.ylim(lims)
plt.show()

```



3

3.1 Regression Model Insights:

Regression analysis quantifies these visual trends:

- Each additional 1 °C increase in temperature $\rightarrow \approx 250$ more rentals per day.
- Each additional 1 mm of rain $\rightarrow \approx 90$ fewer rentals per day.
- On rainy days, rentals drop by an average of $\sim 1,500$ trips.
- After controlling for day-of-week, temperature and rain remain statistically significant, while weekdays (Tue–Thu) see more trips than weekends.

3.2 Model Validation:

The “Predicted vs Actual” scatterplot shows predictions closely aligned with real observations along the 45° line. While there is natural spread ($R^2 \approx 0.17$), the model successfully captures the general pattern: warmer, dry weekdays drive higher ridership.

3.3 Overall Conclusion:

San Francisco’s narrow but consistent temperature range still meaningfully influences daily bike rentals. Warmer and dry days encourage more rides, while rain sharply decreases demand. Temporal factors like weekday and season amplify these effects, demonstrating how even mild weather variability shapes urban mobility behavior.

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